Python Training 2026

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Python Training Program

Python Programming – Upskill Program

Day 1: Introduction to Python & Git Basics



What We Will Do Today

By the end of Day 1, you will:

- Understand what Python is and where it is used
- Set up Python and VS Code
- Write and run your first Python program
- Learn Python syntax, indentation, variables, and data types
- Understand Git & GitHub basics
- Clone a repository and push your first code

What is Python?

Python is a high-level, interpreted programming language created by Guido van Rossum and first released in 1991.

Key characteristics:

- Simple and readable syntax
- Beginner-friendly
- Open-source
- Cross-platform
- Large community support

Python was designed to be easy to learn and fun to use, inspired by Monty Python's Flying Circus.

Van Rossum built Python from scratch in three months using a C compiler on his home Macintosh.



Why Learn Python?

- Easy to read and write
- Faster development compared to other languages
- Used in multiple domains
- Strong community and libraries
- High demand in the job market

Python is often the first programming language recommended for beginners.

Applications of Python

Python is widely used in:

- Web Development (Django, Flask)
- Data Science & Data Analysis
- Machine Learning & Artificial Intelligence
- Automation & Scripting
- DevOps & Cloud
- IoT & Embedded Systems
- Cybersecurity
- Game Development

Python Ecosystem

Python ecosystem includes tools and libraries for:

- **Web** → Django, Flask
- **Data Science** → Pandas, NumPy
- **Visualization** → Matplotlib, Seaborn
- **Machine Learning** → scikit-learn, TensorFlow
- **Automation** → os, shutil, selenium

Python is not one tool, it’s an entire ecosystem.

Companies Using Python

Top companies using Python:

- **Google** – data processing, ML, automation
- **Netflix** – recommendation systems, analytics
- **Instagram (Meta)** – backend & performance optimization
- **Spotify** – data analysis
- **Reddit** – backend services

Python is industry-proven.



Installing Python

1) Download Python from https://www.python.org/downloads/
2) Install Python (check “Add Python to PATH”)
3) Verify installation:

```
python --version
```

Python comes with:

- Python Interpreter
- Standard Library
- pip (package manager)



Installing VS Code

VS Code is a lightweight IDE with:

- Python support
- Debugging tools
- Git & GitHub integration
- Extensions support

Steps:

- Download and Install VS Code from https://code.visualstudio.com/
- Install Python Extension
- (Optional) Install GitHub extension

Running Your First Python Program

```c
//program in C

#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    return 0;
}
```

```cpp
//program in C++

#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

```python
//program in Python

print("Hello, World!")
```

Why Python?

- No boilerplate code
- Easy to read
- Fast execution for beginners

Python Syntax Basics

Syntax refers to the guidelines that determine the structure of a language. In computer programming, syntax specifies the rules governing the arrangement of symbols, punctuation, and words within a programming language.

We Know that programming languages function on the same principles.

If the syntax of a language is not followed, the code will not be understood by a compiler or interpreter.

Key Features of Python Syntax

- **Indentation:** Python uses indentation to define code blocks instead of braces { }. It is mandatory, not optional.
- **Case Sensitivity:** Python is case-sensitive.
- **Comments:** Single line comment (#) , Multiline comment (''' ''').
- **Statements:** Python executes the statement in line by line manner. I.e one line at a time.
- **Input/Output:** input() is used to take user input. And print() is used to show the output.

Variable

Variable is something that represent some information(value) that may be string, number or any data type. They provide a way of labeling data with a descriptive name, so our programs can be understood more clearly by the reader and ourselves.

Naming a variable is important task in programming, so make sure that the name you assign your variable is accurately descriptive and understandable to another reader.

Rules for creating the variable name:

- Must start with a letter or the underscore character
- Cannot start with a number
- Can only contain alphanumeric characters and underscores (A-z, 0-9, and _)
- Case-sensitive (mobile, Mobile and MOBILE are three different variables)
- Cannot be any of the Python keywords. (Check >> help("keywords"))

Python do not have any command for creating the variable, you can just type variable name and the value along with it. Variables do not require a specific type declaration and can change their type even after being assigned a value. Type casting is also there if you want to specify the variable type.

Data Type

Data type refers to the classification or category of data that determines the kind of value a variable can hold and the operations that can be performed on it. It defines the nature of data, such as numeric, textual, or logical, and helps the compiler or interpreter understand how the data will be used.

They define how memory is allocated for the data and also help to ensure program correctness by preventing invalid operations.

Python is a dynamically typed language, meaning variables do not need explicit type declarations.

Data Types	Classes	Description
Numeric	int, float, complex	holds numeric values
String	str	holds sequence of characters
Sequence	list, tuple, range	holds collection of items
Mapping	dict	holds data in key-value pair form
Boolean	bool	holds either `True` or `False`
Set	set, frozenset	hold collection of unique items

What is Git?

Git is a version control system used to:

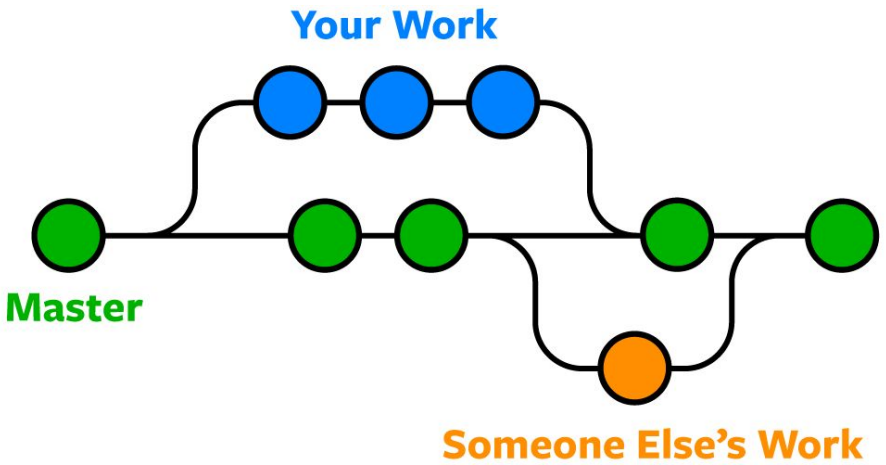
- Track code changes
- Collaborate with teams
- Maintain history of code
- Roll back to previous versions

Git works locally on your computer.



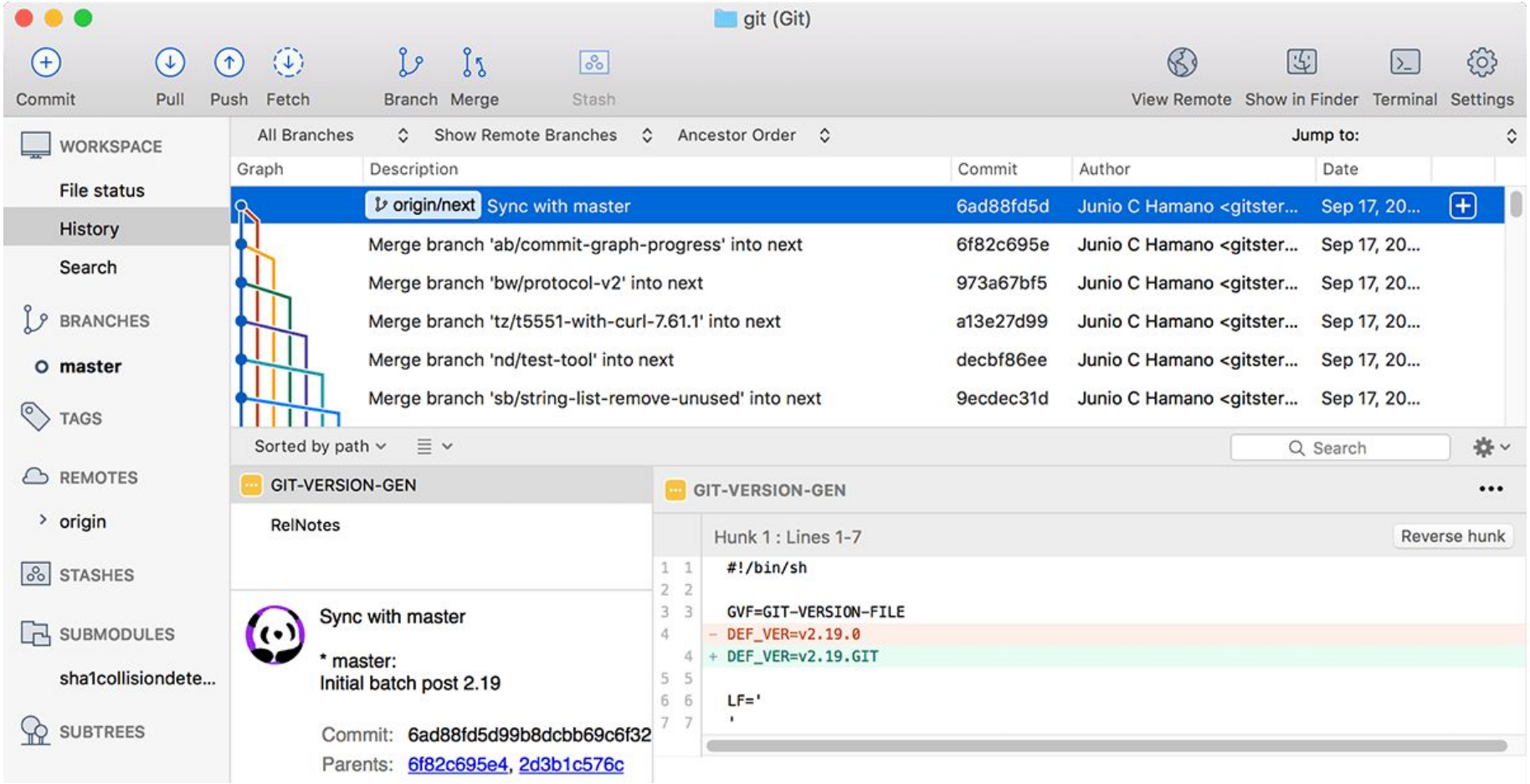
About Git

- Git is the most commonly used version control system.
- Git tracks the changes you make to files, so you have a record of what has been done, and you can revert to specific versions should you ever need to.
- Git also makes collaboration easier, allowing changes by multiple people to all be merged into one source.
- So regardless of whether you write code that only you will see, or work as part of a team, Git will be useful for you.



About Git Contd.

- Git is software that runs locally.
- Your files and their history are stored on your computer.
- You can also use online hosts (such as GitHub or Bitbucket) to store a copy of the files and their revision history.
- Having a centrally located place where you can upload your changes and download changes from others, enable you to collaborate more easily with other developers.
- Git can automatically merge the changes, so two people can even work on different parts of the same file and later merge those changes without losing each other's work!
- Git is software that you can access via a command line (terminal), or a desktop app that has a GUI (graphical user interface) such as Sourcetree shown in the next slide.


git (Git)
Commit
Pull
Push
Fetch
Branch
Merge
Stash
View Remote
Show in Finder
Terminal
Settings
WORKSPACE
File status
History
Search
BRANCHES
master
TAGS
REMOTES
origin
STASHES
SUBMODULES
sha1collisiondete...
SUBTREES
All Branches
Show Remote Branches
Ancestor Order
Jump to:
Graph
Description
Commit
Author
Date
origin/next Sync with master 6ad88fd5d Junio C Hamano <gitster... Sep 17, 20...
Merge branch 'ab/commit-graph-progress' into next 6f82c695e Junio C Hamano <gitster... Sep 17, 20...
Merge branch 'bw/protocol-v2' into next 973a67bf5 Junio C Hamano <gitster... Sep 17, 20...
Merge branch 'tz/t5551-with-curl-7.61.1' into next a13e27d99 Junio C Hamano <gitster... Sep 17, 20...
Merge branch 'nd/test-tool' into next decbf86ee Junio C Hamano <gitster... Sep 17, 20...
Merge branch 'sb/string-list-remove-unused' into next 9ecdec31d Junio C Hamano <gitster... Sep 17, 20...
Sorted by path
Search
GIT-VERSION-GEN
RelNotes
Sync with master
* master:
Initial batch post 2.19
Commit: 6ad88fd5d99b8dcbb69c6f32
Parents: 6f82c695e4, 2d3b1c576c
GIT-VERSION-GEN
Hunk 1 : Lines 1-7
Reverse hunk
#!/bin/sh
GVF=GIT-VERSION-FILE
- DEF_VER=v2.19.0
+ DEF_VER=v2.19.GIT
LF='
'

Git Repositories

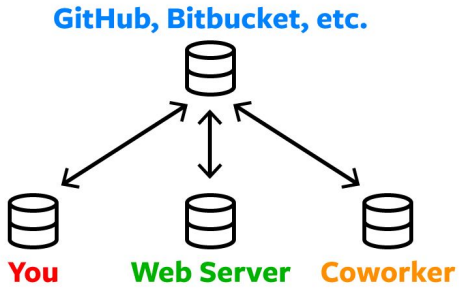
- A Git repository (or repo for short) contains all of the project files and the entire revision history.
- You'll take an ordinary folder of files (such as a website's root folder), and tell Git to make it a repository.
- This creates a .Git subfolder, which contains all of the Git metadata for tracking changes.
- On Unix-based operating systems such as macOS/Windows, files and folders that start with a period (.) are hidden, so you will not see the .Git folder in the macOS Finder/ Windows File Explorer unless you show hidden files, but it's there!
- You might be able to see it in some code editors.

Stage & Commit Files

- Think of Git as keeping a list of changes to files. So how do we tell Git to record our changes? Each recorded change to a file or set of files is called a **commit**.
- Before we make a commit, we must tell Git what files we want to commit.
- This is called staging and uses the add command. Why must we do this? Why can't we just commit the file directly?
- Let's say you're working on a two files, but only one of them is ready to commit. You don't want to be forced to commit both files, just the one that's ready. That's where Git's add command comes in. We add files to a staging area, and then we commit the files that have been staged.

Remote Repositories (on GitHub & Bitbucket)

- Storing a copy of your Git repo with an online host (such as GitHub or Bitbucket) gives you a centrally located place where you can upload your changes and download changes from others, letting you collaborate more easily with other developers.
- After you have a remote repository set up, you upload (push) your files and revision history to it.
- After someone else makes changes to a remote repo, you can download (pull) their changes into your local repo.



What is GitHub?

GitHub is a cloud platform that hosts Git repositories.

GitHub allows:

- Code sharing
- Collaboration
- Pull Requests
- Code reviews
- Portfolio building

Git ≠ GitHub

Git is a tool, GitHub is a platform.



Virtual Environment

- Different projects may require different versions of the same library (e.g., Project A needs an older library version, while Project B needs a newer one).
- It prevents accidental modification or breaking of your global system Python installation.
- A Python virtual environment is a self-contained directory tree that contains a specific Python installation and a separate set of installed packages, which is isolated from other projects and the system-wide Python environment.

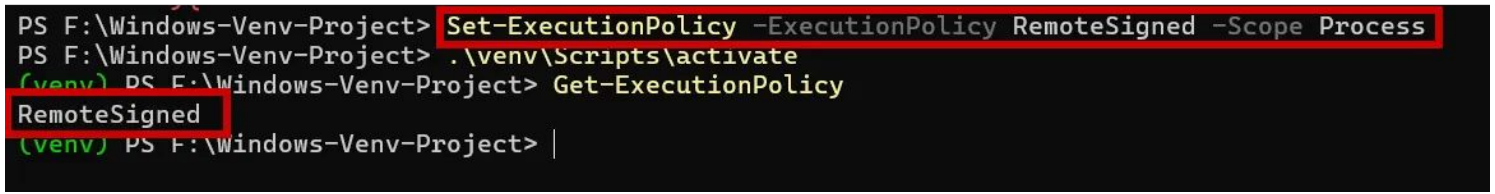
- Create the Environment:
  - Windows/macOS/Linux: *python -m venv venv*
  - Note: On some Linux systems, you might need to run *sudo apt install python3-venv* first if it is not included with your Python installation.
- Activate the Environment:
  - Windows: *venv\Scripts\activate*
  - macOS/Linux: *source venv/bin/activate*
  - Windows Bash: *venv\Scripts\activate.bat*
  - Windows Powershell: *venv\Scripts\Activate.ps1*
- Install Packages:
  - *pip install package_name*
- To share packages:
  - *pip freeze > requirements.txt*
- Deactivate the Environment
  - *deactivate*
- Deleting the Environment
  - macOS/Linux: *rm -rf venv*
  - Windows: *rmdir /s /q venv*

Deactivate it first if it is active

Solving the Activate.ps1 cannot be loaded error on Windows




```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```



What Are Operators?

Operators are symbols that perform operations on variables and values.

Example:

a = 10

b = 5

print(a + b)

Operators tell Python what action to perform.

Arithmetic Operators

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
//	Floor Division
%	Modulus
**	Power

```
print(10 / 3)
print(10 // 3)
print(10 % 3)
```

Comparison Operators

Operator	Meaning
==	Equal
!=	Not equal
>	Greater than
<	Less than
>=	Greater or equal
<=	Less or equal

```
print(5 > 3)
print(10 == 5)
```

These return True or False.

Logical Operators

Used to combine conditions.

Operator	Meaning
and	Both must be True
or	At least one True
not	Reverse condition

```
age = 20
print(age > 18 and age < 30)
```

Type Casting

Type casting converts one data type to another.

```
age = int(input("Enter age: "))
price = float(input("Enter price: "))
```

Common conversions:

- int()
- float()
- str()
- bool()

CONDITIONAL STATEMENTS

Why Conditions?

Conditions allow programs to:

- Make decisions
- Control program flow
- Execute code based on logic

if Statement

```
age = 18

if age >= 18:

    print("You are eligible to vote")
```

Indentation is Mandatory!!!!

if-else Statement

```
age = 16

if age >= 18:

   print("Adult")

else:

   print("Minor")
```

if-elif-else

```
marks = 75
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Grade C")
```

Nested Conditions

```
age = 20

citizen = True

if age >= 18:

  if citizen:

    print("Eligible")
```

Common Beginner Mistakes

- Forgetting indentation
- Using = instead of ==
- Not converting input to int
- Incorrect logical operator usage

Day 4 Overview

- Why loops are needed
- for loop
- while loop
- break, continue, pass
- Real-world loop problems
- Mini practice exercises

Why Do We Need Loops?

Without loops:

```
print("Hello")

print("Hello")

print("Hello")
```

With loop:

```
for i in range(3):

  print("Hello")
```

Loops help us:

- Avoid repetition
- Process large data
- Automate tasks
- Write efficient code

